

2026Fall Quiz

for (MATH261) Discrete Mathematics

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Problem 1.1. Let $X = \mathbb{R}$ and let $R = \emptyset$ be the void relation on X . Determine whether R is reflexive, symmetric, and transitive, and hence whether R is an equivalence relation. Justify each answer.

Solution. The relation R is not reflexive, since $(x, x) \notin R$ for every $x \in X$.

The relation R is symmetric: the implication $xRy \Rightarrow yRx$ is vacuously true because xRy never occurs. Likewise, R is transitive because the premise $xRy \wedge yRz$ never occurs, so

$$(xRy \wedge yRz) \Rightarrow xRz$$

is vacuously true.

Thus R is symmetric and transitive but not reflexive. Therefore R is not an equivalence relation.

□

Problem 1.2. For $n \in \mathbb{N}_0$, prove that

$$\sum_{i=0}^n \binom{n}{i}^2 = \binom{2n}{n}.$$

Solution 1. Let U and V be disjoint sets with $|U| = |V| = n$. Count the n -element subsets of $U \cup V$ according to the number i of selected elements from U . If exactly i elements are chosen from U , then $n - i$ elements must be chosen from V , giving

$$\binom{n}{i} \binom{n}{n-i} = \binom{n}{i}^2$$

choices. Summing over $i = 0, \dots, n$ counts all n -element subsets of the $2n$ -element set $U \cup V$. Hence

$$\sum_{i=0}^n \binom{n}{i}^2 = \binom{2n}{n}.$$

□

Solution 2. By symmetry of binomial coefficients and Vandermonde's identity,

$$\sum_{i=0}^n \binom{n}{i}^2 = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} = \binom{2n}{n}.$$

□